

Phongphat Bangkha

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Summary

Third-year Computer Science student seeking a Cooperative Education opportunity, with a keen interest in **Data Science** and **AI**. Built on a solid foundation in **Software Engineering** and **Web Development**. A responsible learner with a strong sense of ownership, committed to executing and completing assigned tasks effectively.

Education

King mongkut's university of technology north bangkok

Expected Graduation: May 2027

Bachelor of Computer Science

Bangkok, Thai

- Honours Scholar — GPA: 3.68
- Courses: Software Engineering, Machine Learning, Artif Intell Software Dev, Data Structure, Design Analysis Of Algor

Projects

OmniMetric | *Source Code*

Next.js | **Python** | **OpenCV** | **AI Model**

- **Status: In Progress** (Capstone Project for ARTIF INTELL SOFTWARE DEV)
- Developing a spatial analysis web app that estimates **real-world distance and direction** from standard 2D images and **360° panoramas**.
- Engineered a dual-stage AI pipeline integrating **YOLO** for object detection and **Depth estimation models** to generate high-fidelity depth maps.
- Implemented **Metric Depth Estimation** to convert relative depth pixels into precise metric units (meters) for digital analysis.

TravelLens | *Source Code*

Tensorflow | **FastAPI** | **ReactJS** | **Supabase**

- Developed a full-stack **Content-Based Image Retrieval (CBIR)** application ("Shazam for travel") that identifies unknown locations by analyzing visual features.
- Engineered a **FastAPI** backend integrating **ResNet50** to generate **2048-dimensional vector embeddings**.
- Implemented **Approximate Nearest Neighbor (ANN)** search using **Supabase (pgvector)** for high-precision visual matching.

QuickBites | *Website* | *Source Code*

ReactJS | **TailwindCSS** | **MongoDB** | **ExpressJS**

- Architected a full-stack food ordering platform using the **MERN stack** with real-time capabilities.
- Implemented **WebSocket (Socket.io)** for bi-directional communication, enabling instant order updates without page reloads.
- Integrated dynamic **QR code functionality** to route customers to specific table sessions automatically.

ML & Neural Networks Showcase | *Source Code*

Python | **Streamlit** | **Scikit-learn** | **Pandas**

- Developed an interactive web dashboard using **Streamlit** to visualize and train Machine Learning models.
- Implemented Neural Networks to classify image datasets including **MNIST** and **Fashion MNIST** with real-time performance metrics.
- Utilized **Pandas** and **NumPy** for data preprocessing and exploration.

Technical Skills

Languages: Python, C/C++, JavaScript, TypeScript, HTML5, CSS

Developer Tools: Git, Docker, VSCode, PyCharm, IntelliJ, WebStorm, Supabase, MongoDB, PostgreSQL

Libraries/Frameworks: TensorFlow, Pandas, Numpy, Socket.io, ReactJS, NextJS, FastAPI